

Question number	Answer	Additional guidance	Mark
5(a)	A description that includes two of the following points: • use a microarray (1) • to detect mRNA (1) • ref to use of cDNA / fluorescent {labels / probes / tags}		(2)

Question number	Answer	Additional guidance	Mark
5(b)(i)	• C is the correct answer A is not the correct answer as histones are not acting as a transcription factor B is not the correct answer as histones do not form pores in the nuclear membrane D is not the correct answer as histones do not produce RNA polymerase.		(1)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	B is the correct answer A is not the correct answer as crossing over does not regulate gene expression C is not the correct answer as post transcriptional changes do not regulate gene expression D is not the correct answer as post translation changes do not regulate gene expression		(1)

Question number	Answer	Additional guidance	Mark
5(c)(i)	An explanation that includes three of the following points: (section of DNA / gene) has methyl groups added to the {cytosine bases / CPG sites} (1) {codons / gene / locus / base sequence / promotor region} deleted (in parental chromosome) (1) {reduces / inhibits} {binding / action} of RNA polymerase (1) {less / no} {transcription / mRNA production} (1) - so {less / no} {protein synthesis / translation} (1)	accept different mRNA (sequence) produced accept {different / non-functional protein} formed / different amino acid sequence formed	(3)

Question number	Answer	Additional guidance	Mark
5(c)(ii)	A description that includes four of the following points: • {isolated / extracted} gene for HGH (1) • cut plasmid using same restriction {enzyme / endonuclease} (1) • gene inserted into plasmid using ligase (enzyme) (1) • plasmid introduced into E coli (1) • E coli is grown in a fermenter and HGH {extracted / purified} (1)	accept {isolate / cut(out)} gene for HGH using restriction enzyme accept extract mRNA, use reverse transcriptase and form cDNA accept sticky ends (of gene and plasmid) joined using ligase accept vector taken up by {E coli / bacterium} accept bacteria given heat shock ignore gene gun	(4)

